Physics of Complex Systems

Lecture and Tutorials – Prof. Dr. Haye Hinrichsen – Maximilian Schäfferer – SS 2026

Bayes formula.
Source: cern.ch

Exercise 2.1: Drunk Driver Statistics (2P)

According to German police statistics, the age of the drunk drivers involved in car accidents with physical injuries are distributed as follows:

Age	18-24	25-34	35-44	45-54	55-64	≥65
%	26.1	22.4	18.9	18.5	8.7	5.4

At *given* age, the percentage of *female* drivers involved in such accidents is fairly low:

Age	18-24	25-34	35-44	45-54	55-64	≥65
%	9.1	11.9	15.2	14.4	12.3	9.2

(a) What is probability of being female in an such an accident? (1P)

(b) Use Bayes formula to compute the probability of having such an accident depending on the age given that the driver is female. Where is the maximum? (1P)

Exercise 2.2: Poisson distribution (6P)

The poisson distribution $P_\lambda(k) = \frac{\lambda^k}{k!}e^{-\lambda}$ can be understood as the limit of the binomial distribution in the case of “rare events”.

(a) Let $p = \lambda/N$ and take $N \to \infty$ while keeping λ and k constant. Show that in this limit we can approximate $(1-p)^{N-k} \approx e^{-\lambda}$. (1P)

(b) Show similarly that $\binom{N}{k} \approx \frac{N^k}{k!}$. (1P)

(c) Use (a) and (b) to show that in this limit, the binomial distribution tends to the Poisson distribution. (1P)

(d) Check that the Poisson distribution is properly normalized. (1P)

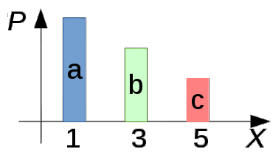
(e) Compute the moment- and cumulant-generating functions (MGF and CGF). (1P)

(f) Determine all cumulants. (1P)

Exercise 2.3: Reconstruct probabilities from given moments (4P)

Consider a system with three configurations $\Omega = \{a, b, c\}$ together with an associated map $X : \Omega \to \mathbb{R}$ defined by

$$X_a = 1, \quad X_b = 3, \quad X_c = 5\,.$$



Please turn over ⇒

(a) How many moments do you have to know in order to be able to reconstruct the probabilitiy distribution $\{P_a, P_b, P_c\}$ and why?

(b) Suppose that M_1 and M_2 are given (with the map X defined above). Find the solution for the probabilities P_a, P_b, and P_c.

(c) Not all possible values for M_1 are M_2 lead to consistent results. Give a counter-example.

(d) Let $M_1 = \frac{7}{2}$ and $M_2 = 15$. Compute the probabilities P_a, P_b, P_c and find a general formula for the higher moments M_n with $n = 3, 4, \ldots \infty$.

$(\Sigma = \mathbf{12P})$

Please submit your solution electronically as a single PDF file until Wednesday, April 29, 2026, via WueCampus.